

Production Server Operations Handbook

This handbook covers routine operations for Engineering production servers: patching, health checks, capacity limits and backups. Changes outside these rules need Engineering Manager approval.

1 Scope

The handbook applies to the web tier, the API tier and the worker tier in the production environment. Staging servers follow the same rules except where a section says otherwise.

2 Patching

Security patches are applied during the approved maintenance window. Emergency patches for actively exploited vulnerabilities may be applied at any time with incident commander approval.

2.1 Web Tier

Web servers are patched in batches of at most 25% of the fleet so that capacity never drops below 75%. Each batch must pass health checks for 10 minutes before the next batch starts.

2.2 API Tier

API servers are patched one server at a time. A server may only be returned to the load balancer after its health check has passed three times in a row. The API tier is never patched during a deployment window.

2.3 Worker Tier

Worker servers are drained before patching: no new jobs are assigned, and running jobs are allowed up to 15 minutes to finish. Jobs still running after 15 minutes are re-queued, not killed.

3 Health Checks

Every production server exposes a health endpoint that is polled every 30 seconds. A server is marked degraded after three consecutive failed checks and offline after ten consecutive failed checks.

A degraded server keeps serving traffic but is flagged on the operations dashboard. An offline server is removed from the load balancer automatically and the owning team is paged.

4 Capacity Limits

Sustained CPU above 85% for 20 minutes on any web or API server triggers an automatic scale-out of two servers. Memory above 90% for 10 minutes triggers a page to the owning team instead, because memory pressure usually indicates a leak rather than load.

Disk usage above 80% raises a warning and above 95% raises a page. Log volumes are rotated daily and kept for 14 days on the server.

5 Backups

Database servers are backed up every 6 hours, and backups are kept for 35 days. A restore test is performed on the first Tuesday of every month; the result is recorded in the change log.

Application servers are not backed up. They are rebuilt from the release artifact and configuration management, which is why no manual change may be made on a production server without also committing it to configuration management.